

DeepSeek-V2

A Strong, Economical, and Efficient MoE Language Model

236B

Total Parameters

21B

Activated / Token

128K tokens

Context Length

Pretrained on 8.1T tokens • SFT + GRPO RL • arXiv: 2405.04434

The LLM Scaling Dilemma: Performance vs. Cost and Inference Speed

Why bigger models create deeper engineering challenges

Training Cost Explosion

- Dense models scale $O(N^2)$ in compute
- Doubling params $\rightarrow 4\times$ cost
- GPT-4 / Llama-70B require thousands of GPU-days
- Limits accessibility for most research labs

KV Cache Bottleneck

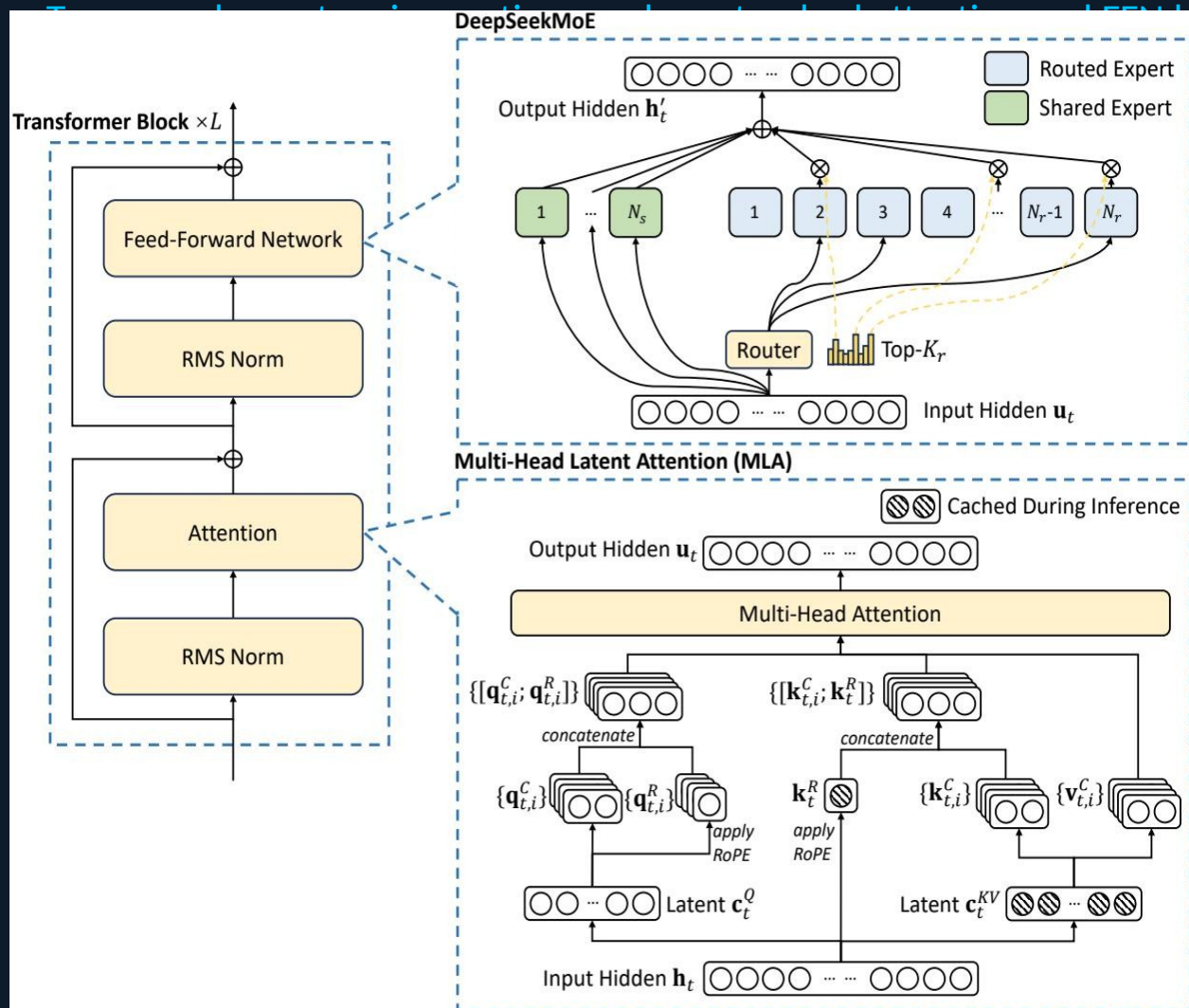
- MHA stores full K,V per head per token
- Cache grows linearly with context length
- Limits batch size & max sequence length
- Dominates memory at inference time

Inference Throughput Gap

- Larger dense models $\rightarrow$ slower generation
- Low throughput $\rightarrow$ high serving cost
- Existing efficiency tricks (GQA, MQA) sacrifice accuracy
- No free lunch in the accuracy-speed trade-off

DeepSeek-V2 answer: MLA (low-rank KV compression) + DeepSeekMoE (sparse experts) $\rightarrow$ 42.5% lower training cost, 93.3% smaller KV cache, 5.76 $\times$ higher throughput

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in a Transformer Block



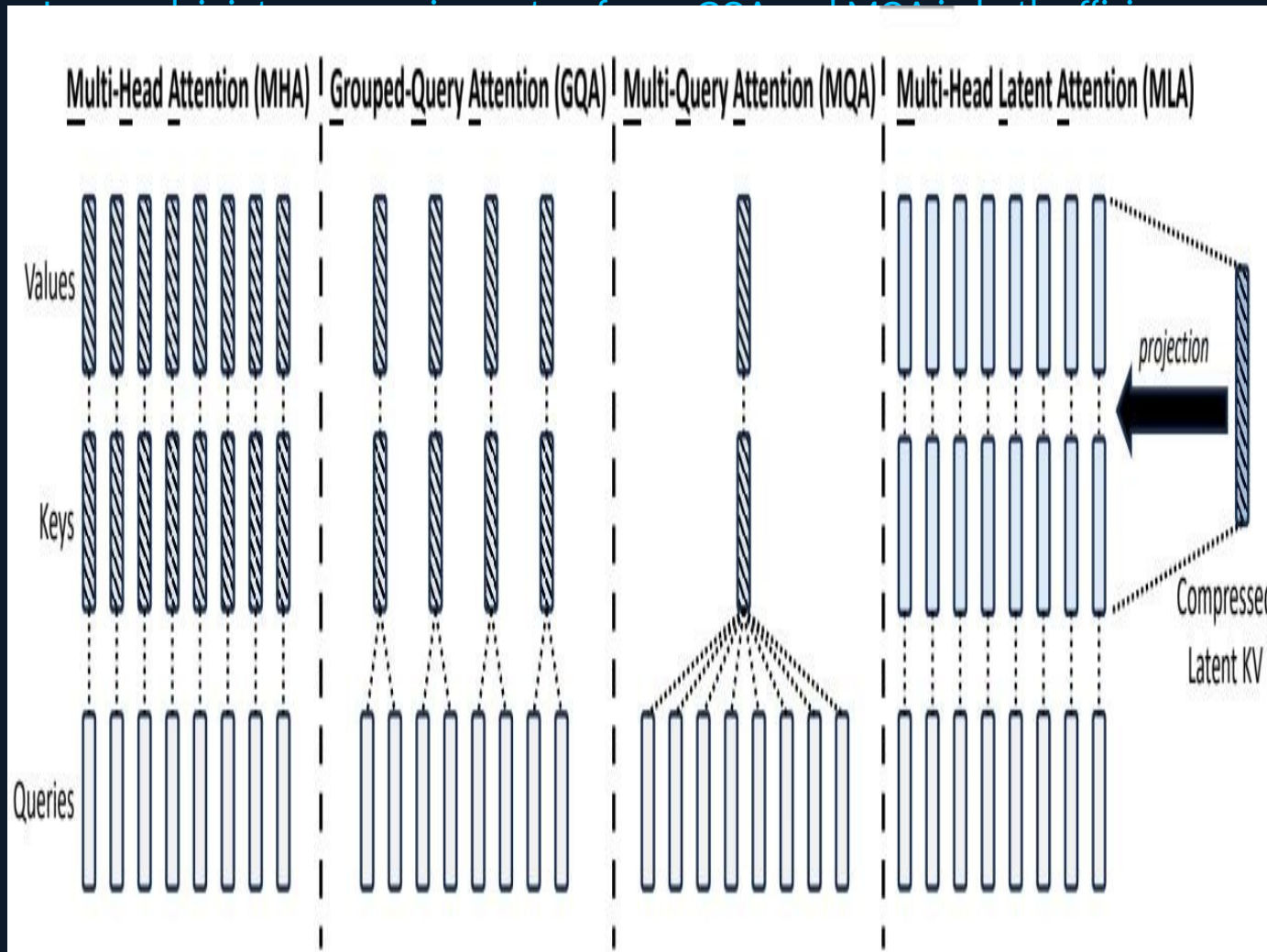
Multi-Head Latent Attention (MLA)

- Replaces Multi-Head Attention (MHA)
- Low-rank joint KV compression
- Decoupled RoPE for positional encoding
- 93.3% KV cache reduction at inference

DeepSeekMoE (FFN replacement)

- Replaces dense Feed-Forward Network
- Shared experts (always active)
- Routed experts (top-K sparse activation)
- Device-limited routing $\rightarrow$ less cross-node comm.

Multi-Head Latent Attention Compresses KV Cache by 93.3%



quality

MHA Problem: Cache full K,V per head $\rightarrow O(n_heads \times d_head)$ per token

GQA / MQA: Reduce heads but still cache raw K,V vectors

MLA Innovation: Compress K,V into compact latent c_t^{kv} via low-rank projection

Reconstruction: Full K,V recovered on-the-fly at inference via learned matrices

Decoupled RoPE: Positional key added separately to preserve length generalization

Result: Superior accuracy vs. MHA with 93.3% smaller cache footprint

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Sparse activation achieves dense model quality at a fraction of the compute

Shared Experts (N_s)

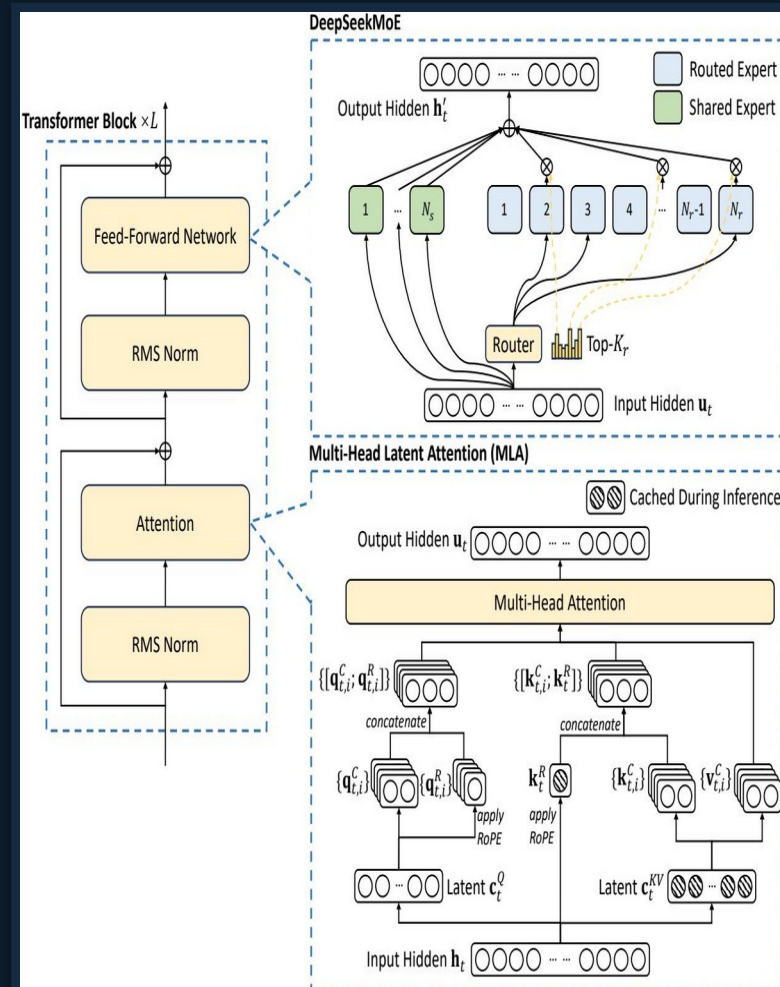
- Always activated regardless of input
- Capture universal, task-agnostic knowledge
- Prevent redundant learning in routed experts
- Stabilize training across diverse data

Routed Experts (N_r)

- Sparsely activated via top- K_r router per token
- Fine-grained segmentation $\rightarrow$ higher specialization
- Device-limited routing: each token uses experts on $\leq M$ devices
- Reduces cross-node communication overhead

Training & Load-Balancing

- Expert-level balance loss: distributes load evenly
- Device-level balance loss: prevents single-device overload
- Token-dropping: prevents expert overflow during training
- No token-dropping at inference $\rightarrow$ full quality retained
- Result: 236B params, only 21B activated per token



42.5% Training Cost Reduction and 5.76× Generation Throughput vs.

DeepSeek 67B

Relative efficiency gains of DeepSeek-V2 over its dense predecessor

DeepSeek 67B → DeepSeek-V2 (Relative Comparison)

-42.5%

Training Cost

vs. DeepSeek 67B dense baseline

-93.3%

KV Cache Size

per token at inference (MLA compression)

5.76×

Generation Throughput

max tokens/sec vs. DeepSeek 67B

MLA → KV Cache

Low-rank joint compression caches compact latent vector instead of full K,V pairs.

MoE → Training Cost

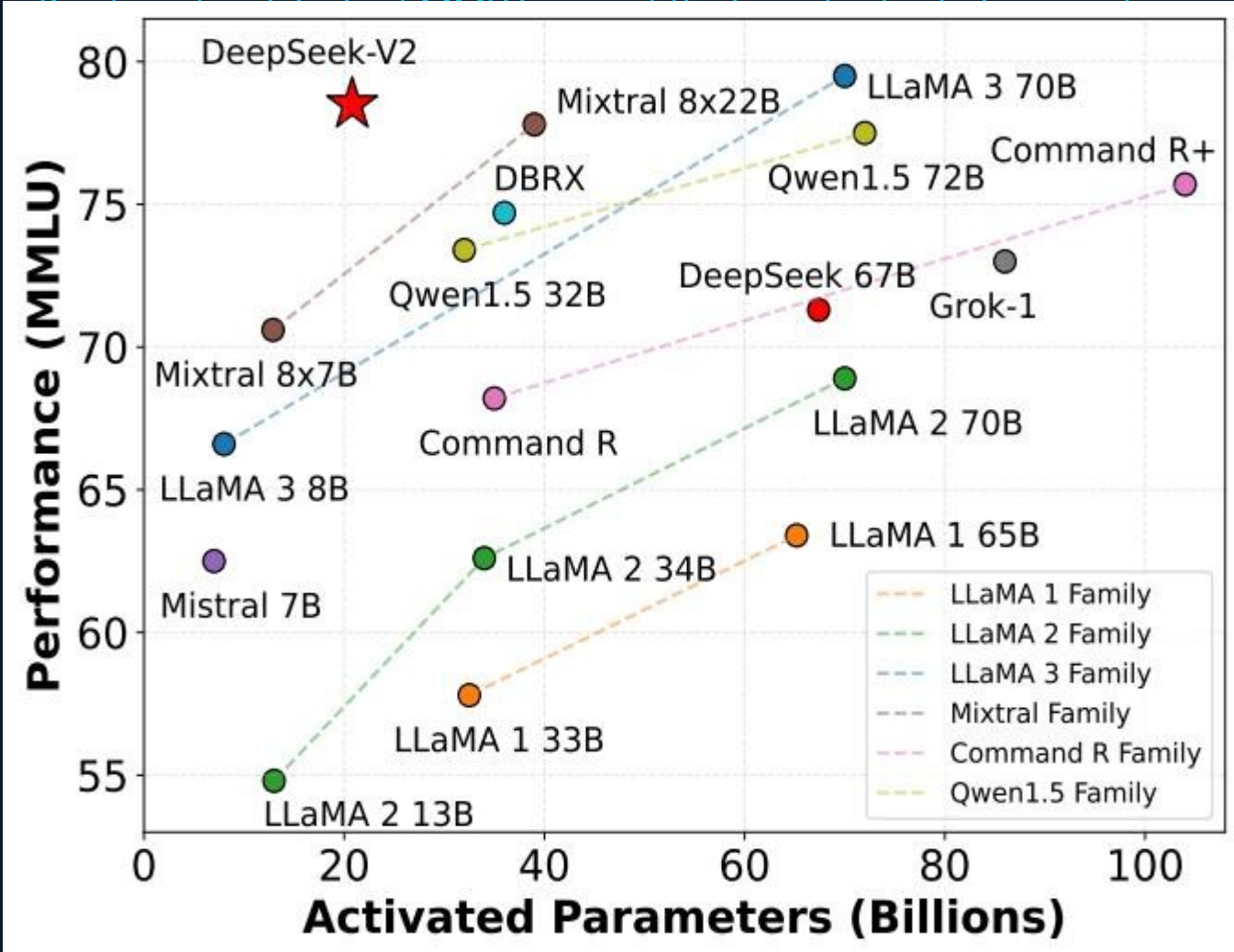
Only 21B of 236B params activate per token; sparse routing cuts total FLOPs.

Both → Throughput

Smaller KV cache enables larger batch sizes; MoE sparsity accelerates decoding.

All metrics are relative comparisons to DeepSeek 67B (dense model) reported in the original paper.

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B Activated Parameters



Model	Act. Params	MMLU
DeepSeek-V2	21B	78.5
Mixtral 8×22B	39B	77.6
LLaMA 3 70B	70B	77.0
Qwen1.5 72B	72B	75.6
DeepSeek 67B	67B	73.0
Mixtral 8×7B	13B	70.6
LLaMA 2 70B	70B	69.0
LLaMA 3 8B	8B	65.0

★ DeepSeek-V2 achieves highest MMLU (78.5) with only 21B activated parameters — 3× fewer than LLaMA 3 70B, yet 1.5 pts higher.

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

SFT + GRPO-based RL alignment unlocks top open-source chat performance

Alignment Pipeline

- Phase 1 – SFT: Fine-tuned on 1.5M high-quality conversational sessions
- Phase 2 – RL: Group Relative Policy Optimization (GRPO) without a separate critic model
 - GRPO uses group-level relative rewards to guide policy updates efficiently

38.9%

AlpacaEval 2.0

Length-controlled win rate vs. GPT-4 (open-ended EN)

8.97

MT-Bench

Multi-turn instruction following, scale 1-10

7.91

AlignBench (ZH)

Comprehensive Chinese alignment benchmark

On AlignBench (Chinese), DeepSeek-V2 Chat (RL) surpasses ALL open-source models and MOST closed-source models — demonstrating state-of-the-art multilingual alignment.

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

1

MLA is a strictly better alternative to MHA

93.3% KV cache reduction with no accuracy loss — GQA/MQA trade accuracy for efficiency; MLA achieves both simultaneously through low-rank latent compression.

2

DeepSeekMoE enables economical training at scale

236B total params, only 21B activated per token. Training cost 42.5% lower than a dense 67B model through sparse expert routing and fine-grained expert segmentation.

3

21B activated params outperform 70B dense models

DeepSeek-V2 (78.5 MMLU) exceeds LLaMA 3 70B (77.0) and Qwen1.5 72B (75.6) with 3× fewer activated parameters — demonstrating MoE efficiency at the frontier.

4

5.76× throughput makes deployment practical

Smaller KV cache → larger inference batches → higher GPU utilization. Real-world deployment cost is dramatically lower than dense 70B alternatives.

5

Top multilingual alignment, fully open-source

DeepSeek-V2 Chat (RL) is the strongest open-source chat model in Chinese and competitive with closed-source systems in English — with weights publicly available.